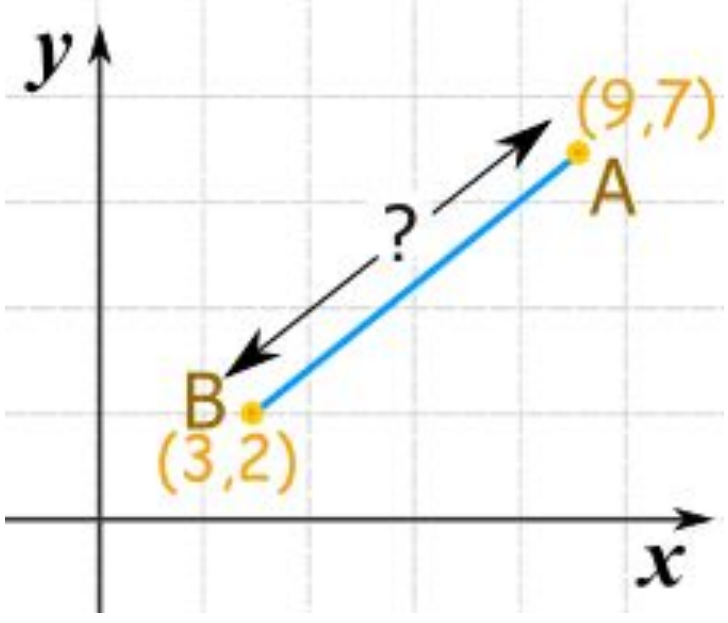
Warm-Up:

What is the distance between A and B?



$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{(9 - 3)^2 + (7 - 2)^2}$$

$$\sqrt{6^2 + 5^2}$$

$$\sqrt{36 + 25}$$

$$\sqrt{61}$$

$$7.8102...$$

Introduction to Unsupervised Learning

K-Means Clustering

Katie Sylvia

Agenda

- Supervised vs. Unsupervised Learning
- Clustering Use Cases & Examples
- Intro to K-Means Clustering
- Pros & Cons
- Creating our first model! (Jupyter Lab)

In Slack:

What are the differences between supervised and unsupervised learning?

Key Differences - Let's Compare to Classification

Supervised

- Input data is labeled.
- Used for prediction.
- Data is classified based on the training dataset.
- Has a feedback mechanism.
- A known number of classes.

Examples: Logistic Regression, Decision Trees, Support Vector Machine

vs.

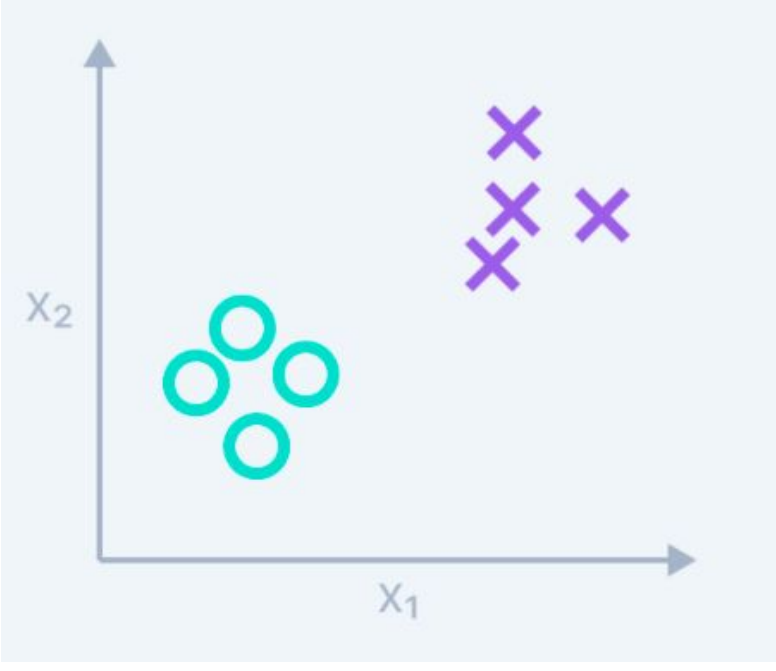
Unsupervised

- Input data is not labeled.
- Used for analysis.
- Assigns properties of given data to classify it. (We do NOT train-test-split).
- Has no feedback mechanism.
- Unknown number of classes.

Examples: K-Means Clustering, Hierarchical Clustering, Apriori Algorithm

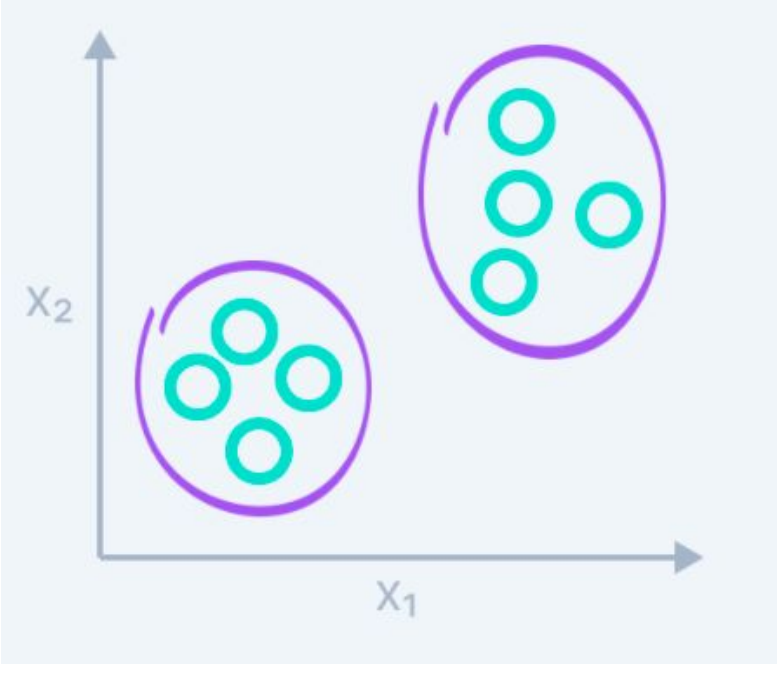
Key Differences - Let's Compare to Classification

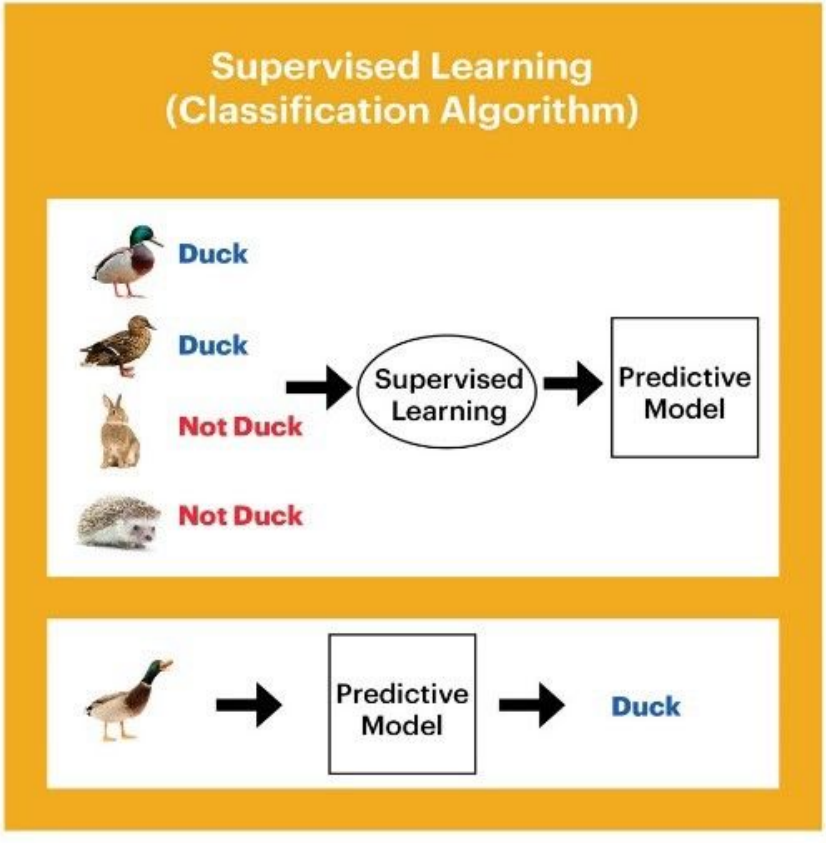
Supervised

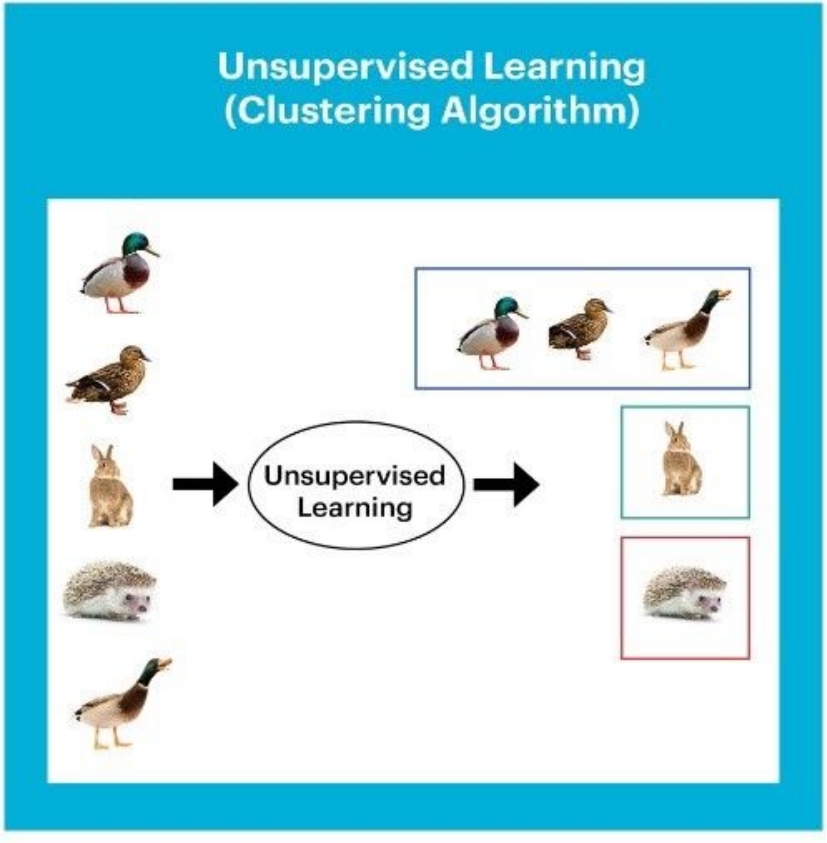


vs.

Unsupervised







Western Digital.

CLASSICAL MACHINE LEARNING

Data is pre-categorized or numerical

Data is not labeled in any way

SUPERVISED

Predict a category

Predict a number

CLASSIFICATION

«Divide the socks by color»



REGRESSION

«Divide the ties by length»



UNSUPERVISED

Divide by similarity

Find hidden dependencies

Identify sequences

CLUSTERING

«Split up similar clothing into stacks»



ASSOCIATION

«Find what clothes I often wear together»



DIMENSION REDUCTION (generalization)

«Make the best outfits from the given clothes»



In Slack:

Can you think of any use cases for clustering?

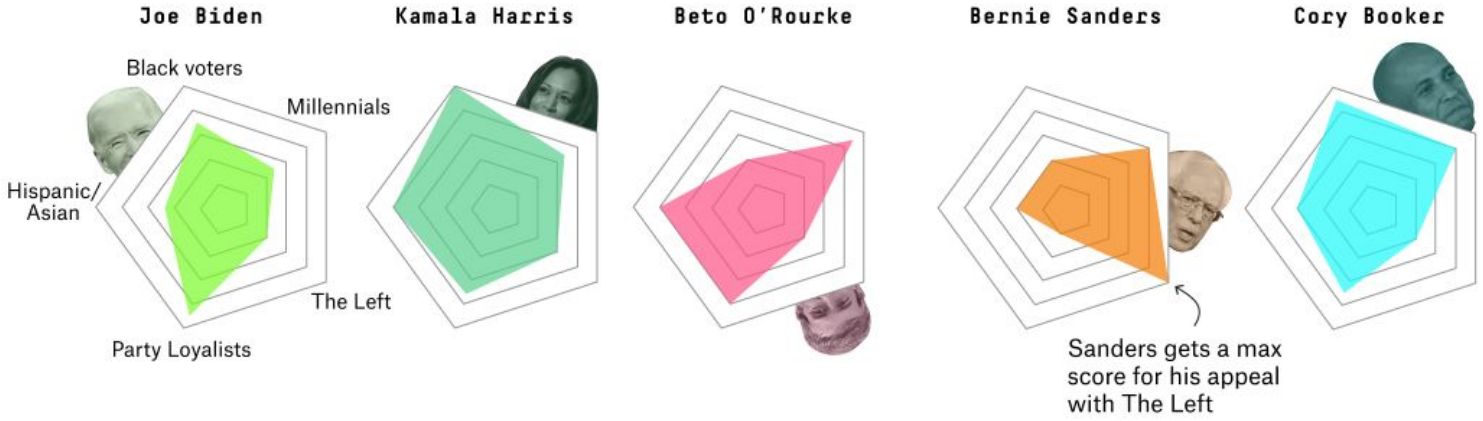
Top 3 Use Cases for Clustering

1. Political Polling

How 2020 contenders rate with 5 corners of the Democratic Party

Mapping the estimated appeal that possible 2020 candidates have with each of five voter groups

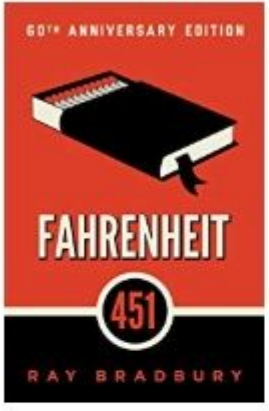




FiveThirtyEight

2. Recommender Systems

Customers who bought this item also bought



Fahrenheit 451
› Ray Bradbury
3,502
#1 Best Seller in Censorship & Politics
Paperback
$8.99



The Glass Castle: A Memoir
› Jeannette Walls
7,651
#1 Best Seller in Journalist Biographies
Paperback
$9.79

In Slack:

What other recommender systems can you think of?

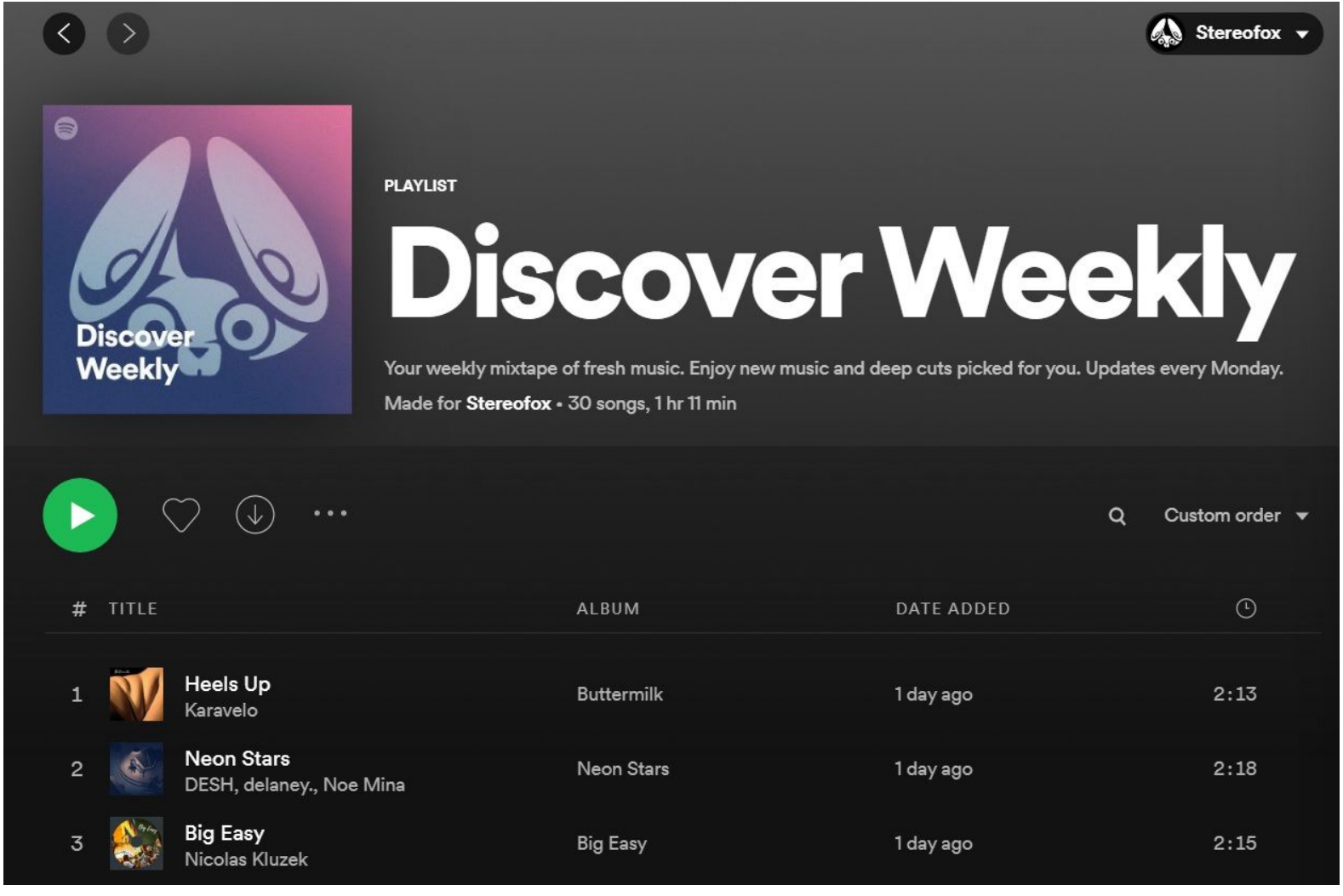
Everything is a Recommendation



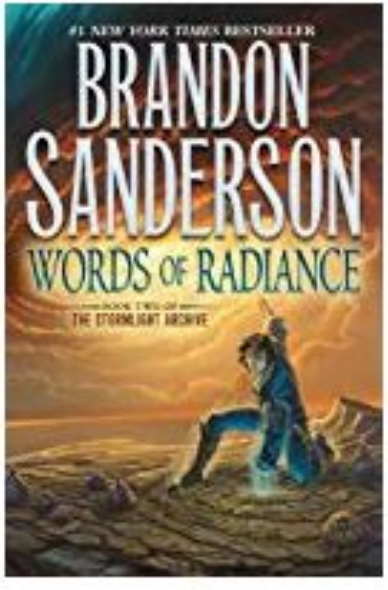
Over 80% of what people watch comes from our recommendations

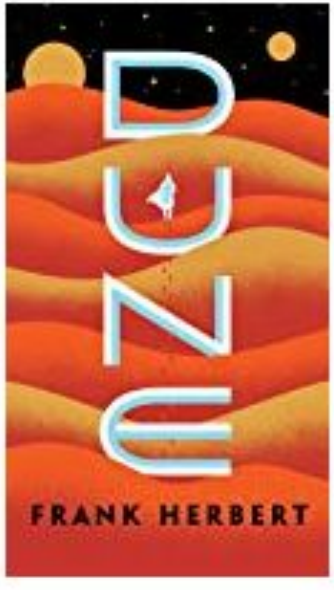
Recommendations are driven by **Machine Learning**

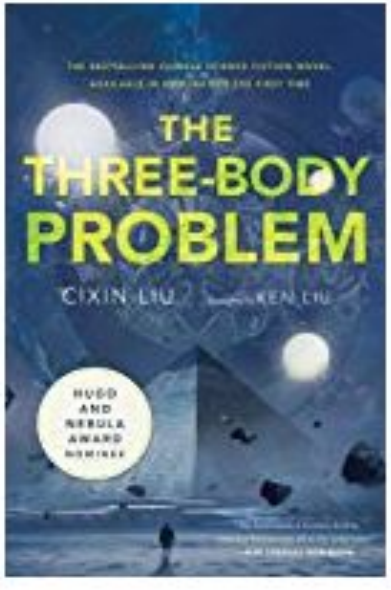
NETFLIX

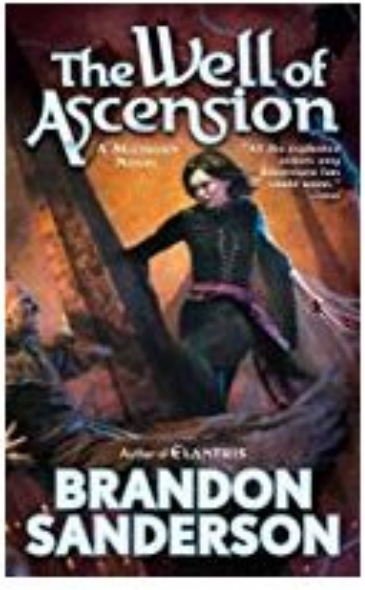

Stereofox
Discover Weekly
PLAYLIST
Discover Weekly
Your weekly mixtape of fresh music. Enjoy new music and deep cuts picked for you. Updates every Monday.
Made for Stereofox • 30 songs, 1 hr 11 min
Custom order
TITLE
ALBUM
DATE ADDED
1 Heels Up Karavelo Buttermilk 1 day ago 2:13
2 Neon Stars DESH, delaney., Noe Mina Neon Stars 1 day ago 2:18
3 Big Easy Nicolas Kluzek Big Easy 1 day ago 2:15

Recommended for You in Kindle Books



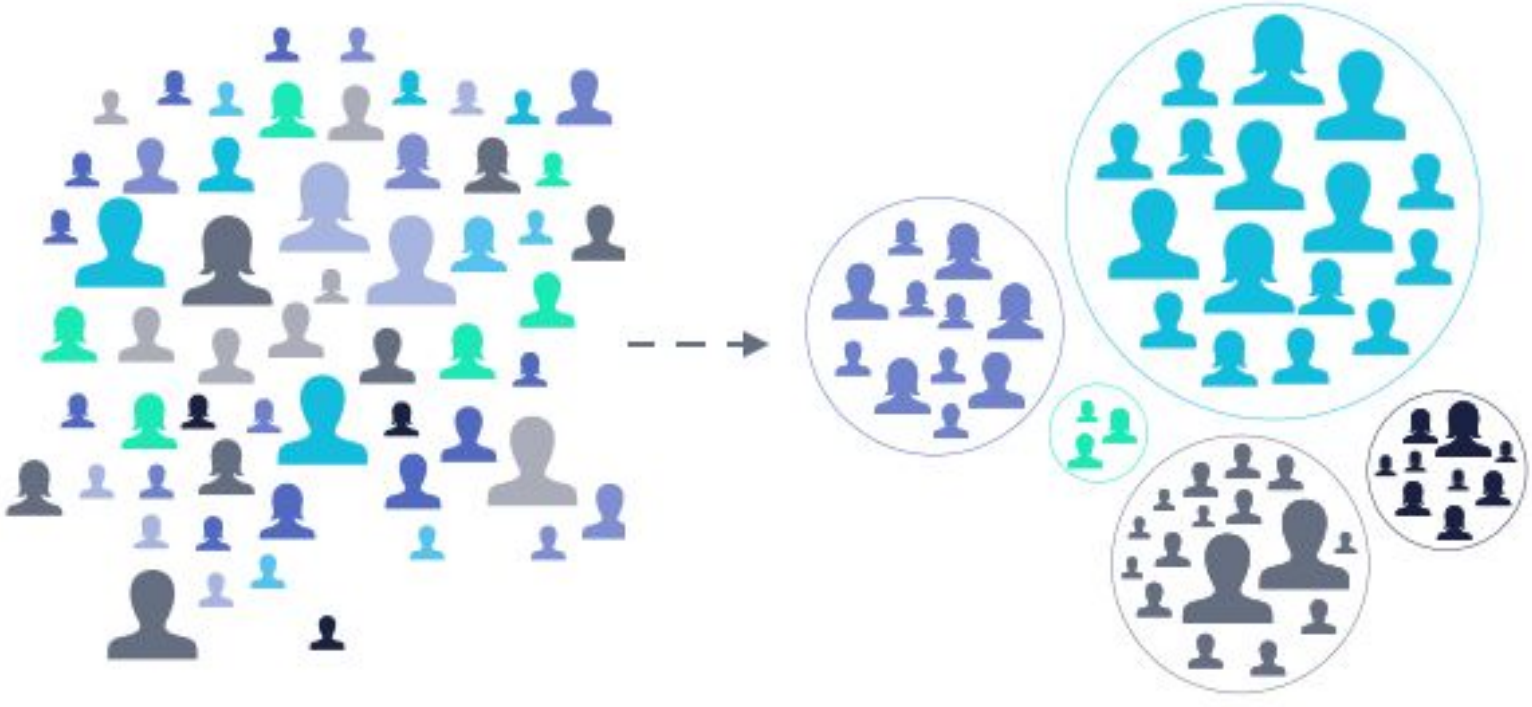






3. Market Research

Customer Segmentation





The WHO

- Age
- Gender
- Ethnicity
- Income
- Level of education
- Religion
- Profession/role in a company



The WHY

- Personality traits
- Hobbies
- Life goals
- Values
- Beliefs
- Lifestyles



The WHERE

- Country
- Region
- City
- Postal code



The HOW

- Spending habits
- Purchasing habits
- Browsing habits
- Interactions with the brand
- Loyalty to brand
- Previous product feedback

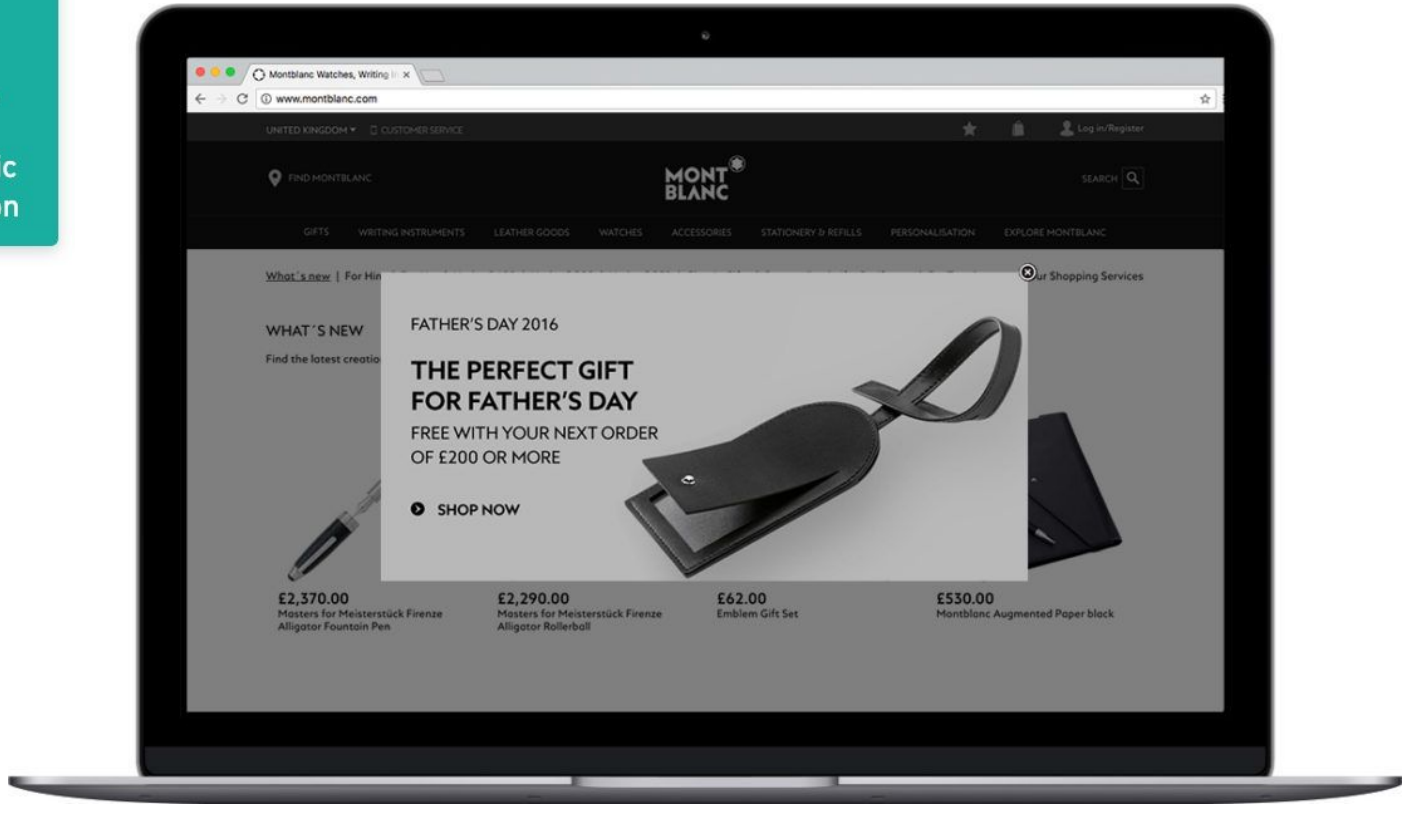
The following examples are pulled from a company called



“Yieldify is a customer journey optimization platform that brings personalization to the full customer journey. It combines award-winning software with a proven methodology to deliver measurable results quickly and easily.

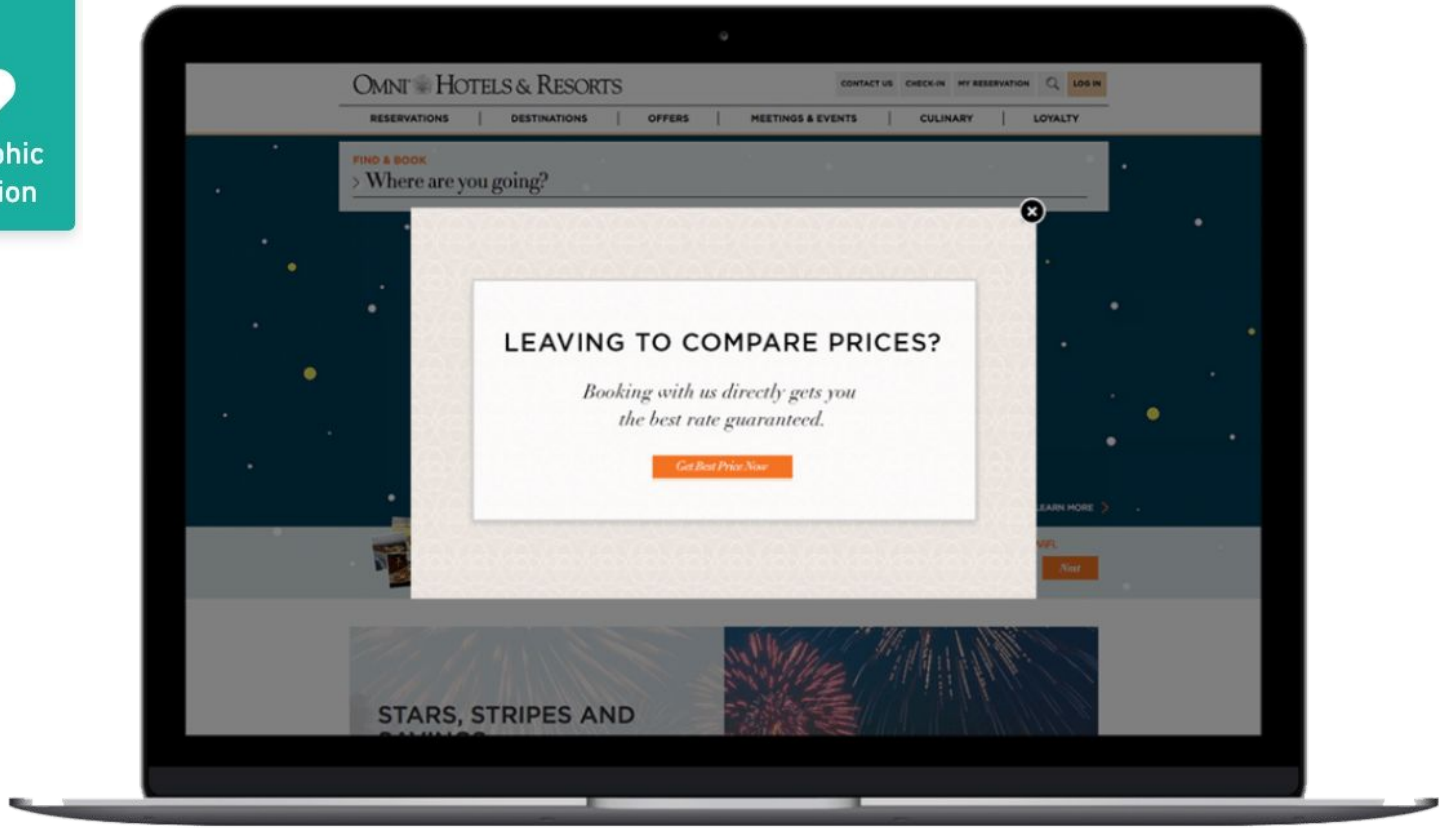
To date, Yieldify has influenced over 100 million sales through more than 200,000 journeys. It currently delivers customer journey optimization for over 1,000 leading e-commerce websites, including Domino’s Pizza, L’Oreal and Megabus.”

Demographic segmentation

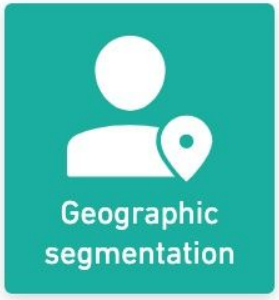


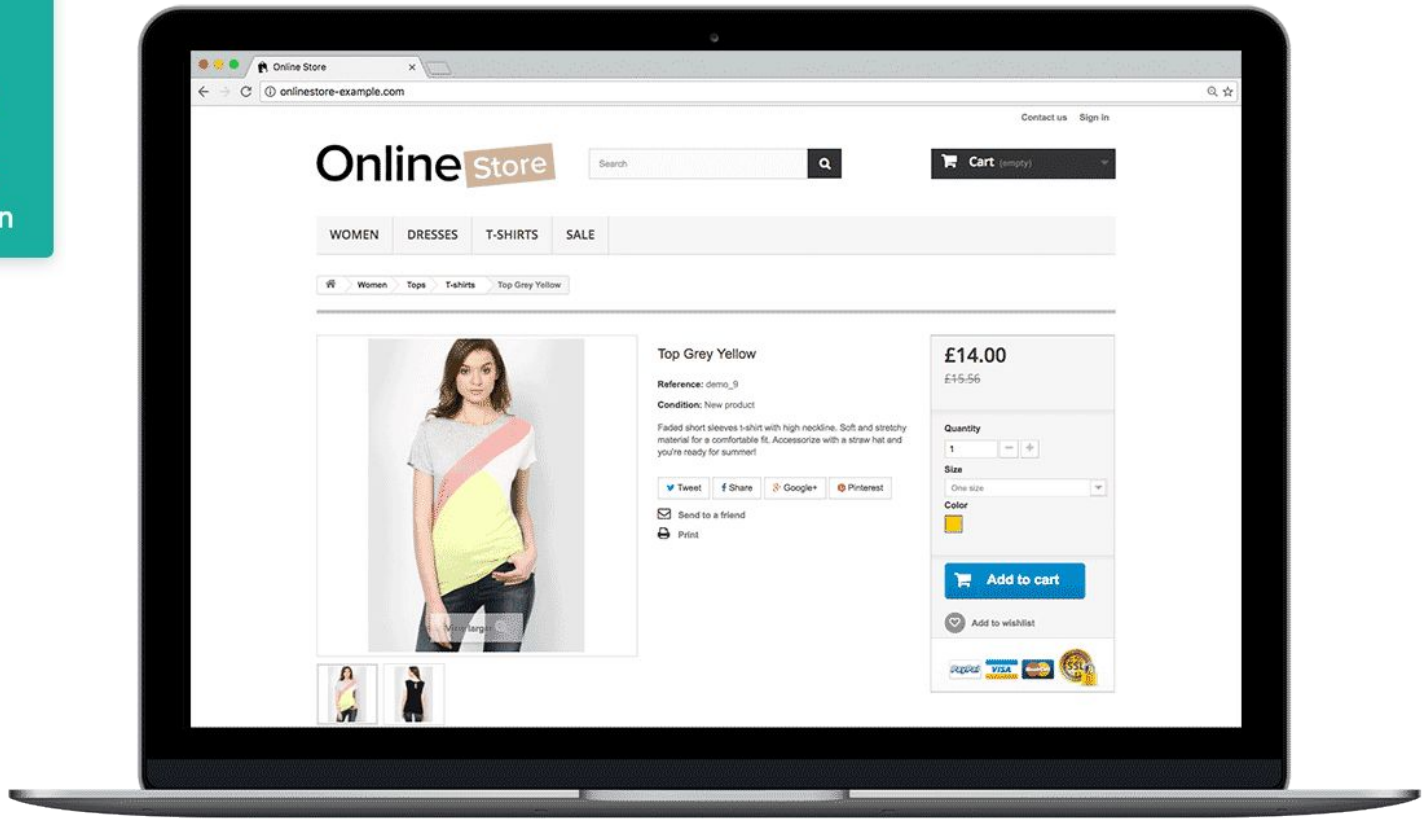
- This Father's Day deal saw a +118% uplift in conversions for those targeted.
- Another offer was aimed specifically at corporate gift buyers – a market segment that Montblanc particularly appeals to – and resulted in a +30% uplift for that segment.

Psychographic segmentation



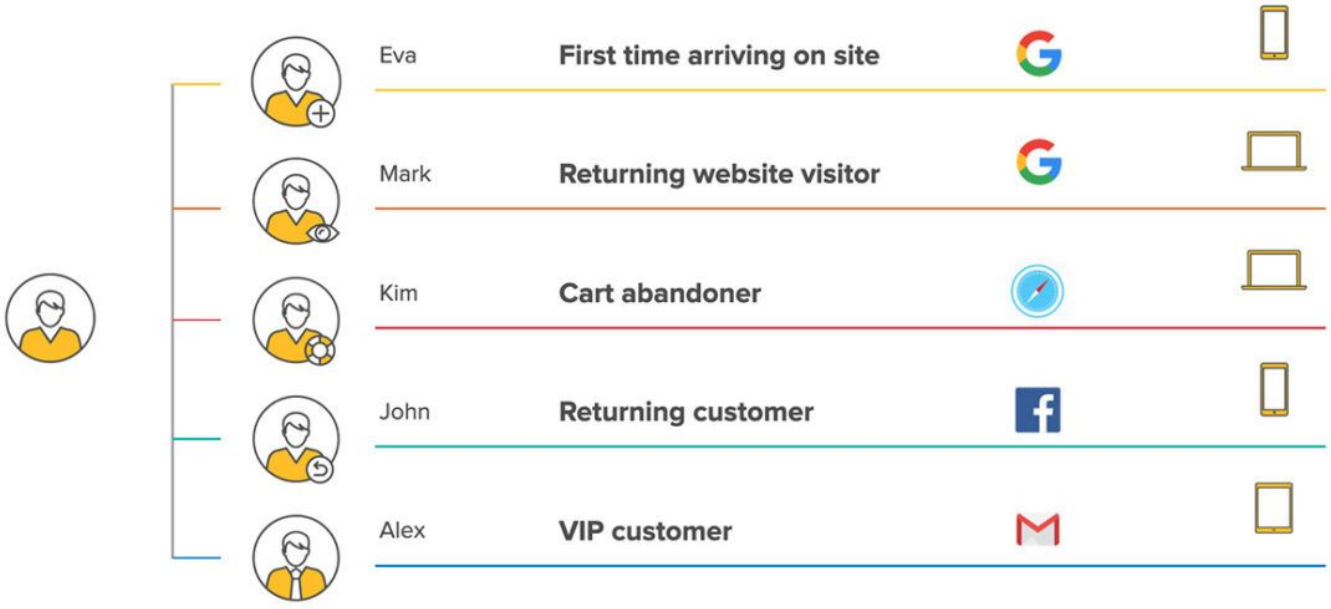
- By targeting a notification campaign specifically towards comparison shoppers, Omni Hotels & Resorts achieved a 39% conversion rate uplift.





- An online shoe store could show different products depending on where the visiting customer was based: rain boots for someone in the countryside, pavement-friendly sneakers for a city-dweller, strappy sandals to resort visitors, and so on!
- Seasonal Segmentation - allows businesses to market season-appropriate products to customers.



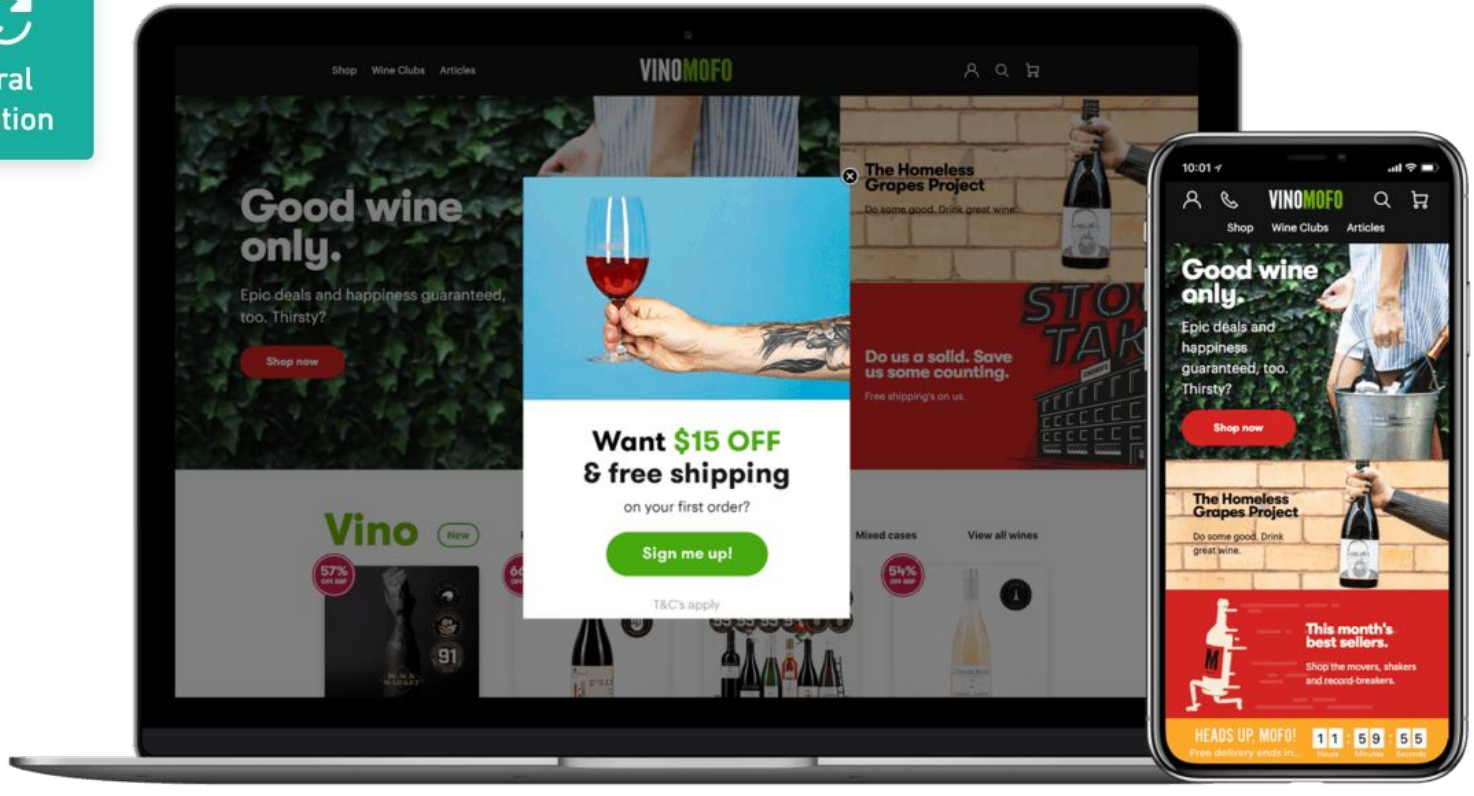


Behavioral Patterns:

- Number of sessions to your website
- Number of pages visited
- Time spent on site
- URLs visited
- Page types visited
- Shopping cart value
- Campaign history
- Referral source
- Exit intent
- Inactivity, and more.

Behavioral segmentation

First time visitor: *Hey, learn about our latest collection!*
Returning visitor: *Join our loyalty program and start saving!*

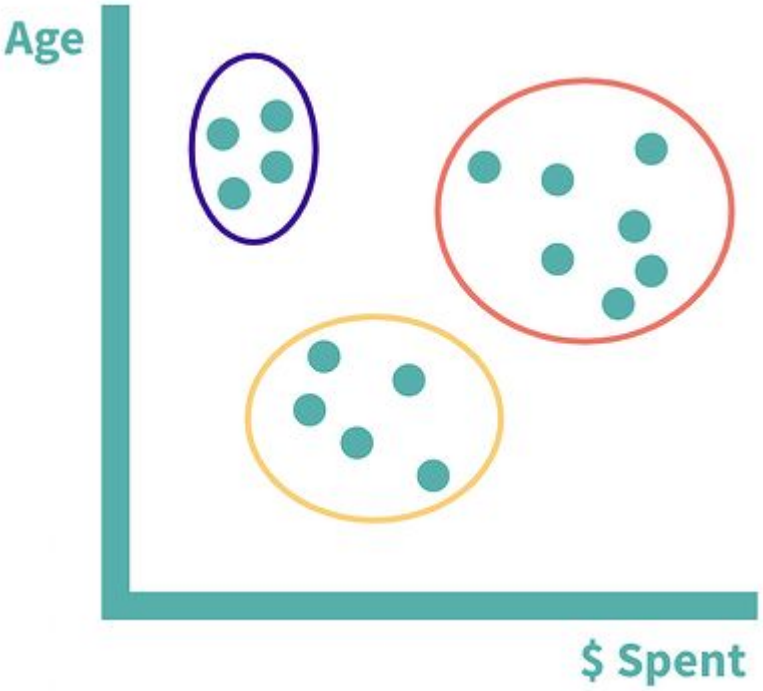


- This targeted approach focused on purchasing habits reached a 34.02% conversion rate uplift with new customers.

Let’s Start Visualizing This

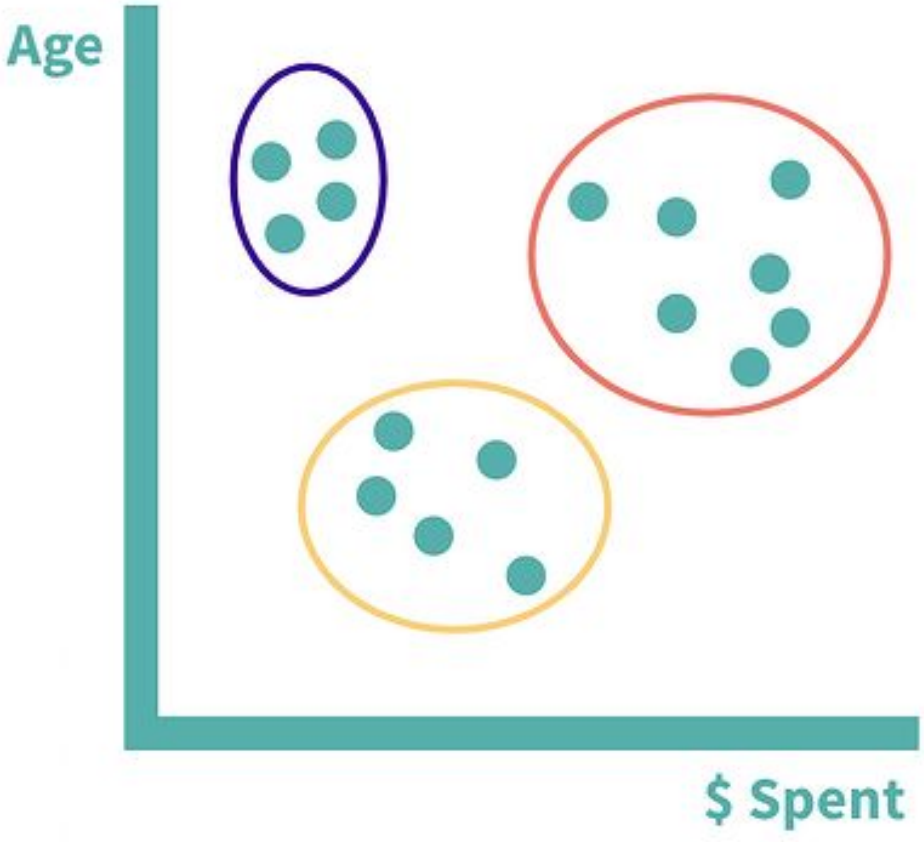


Let's pretend we're the owners of Laffy Taffy Candy Store, and we want to perform customer segmentation. For each customer, we just have two data points: age and dollars spent at our store last week.



Clustering will enable us to analyze the groups of consumers that purchase from our store, and then market more appropriately to each customer group.

What are we noticing?



We were able to visualize these clusters with our own eyes. But what if we cannot? What if we had 3 features? 4? 5? More?

How can we cluster using a machine learning model?

Introduction to K-Means Clustering

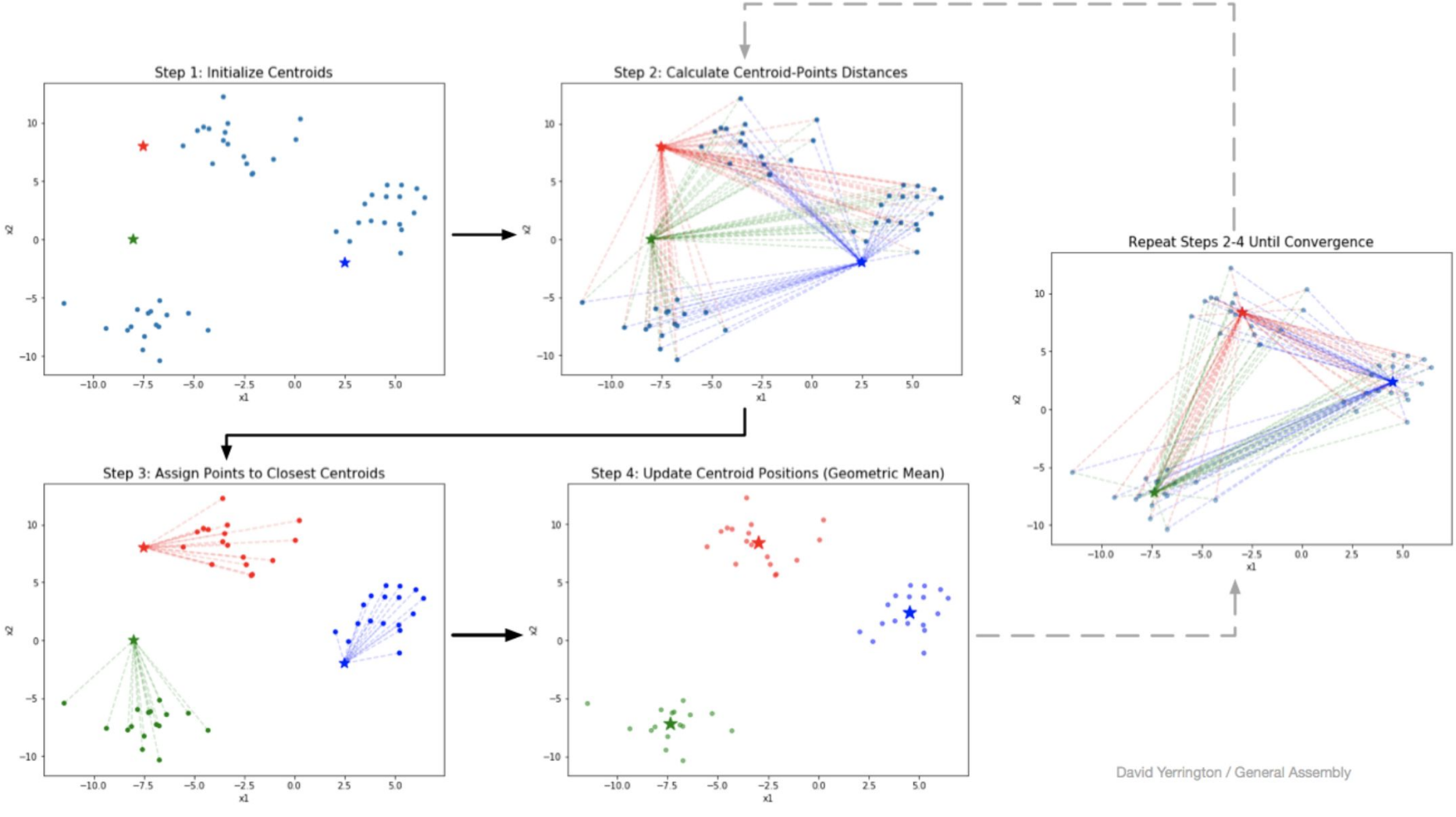
K-Means is the most popular clustering algorithm.

- **k** is the number of clusters (we determine this).
- **Means** refers to the mean points of the k clusters.

Let's go visualize this!

K-Means Clustering Steps:

1. Pick a value for k (the number of clusters to create).
2. Initialize k 'centroids' (starting points). More on this on the later!
3. Create clusters by assigning each data point to its nearest centroid.
4. Make your clusters better. Reassign each centroid to the center of its cluster.
5. Repeat steps 3-4 until the centroids converge and do not change across iterations.


Step 1: Initialize Centroids
Step 2: Calculate Centroid-Points Distances
Repeat Steps 2-4 Until Convergence
Step 3: Assign Points to Closest Centroids
Step 4: Update Centroid Positions (Geometric Mean)
x1
x2
David Yerrington / General Assembly

So... how are centroids initialized? (Three options)

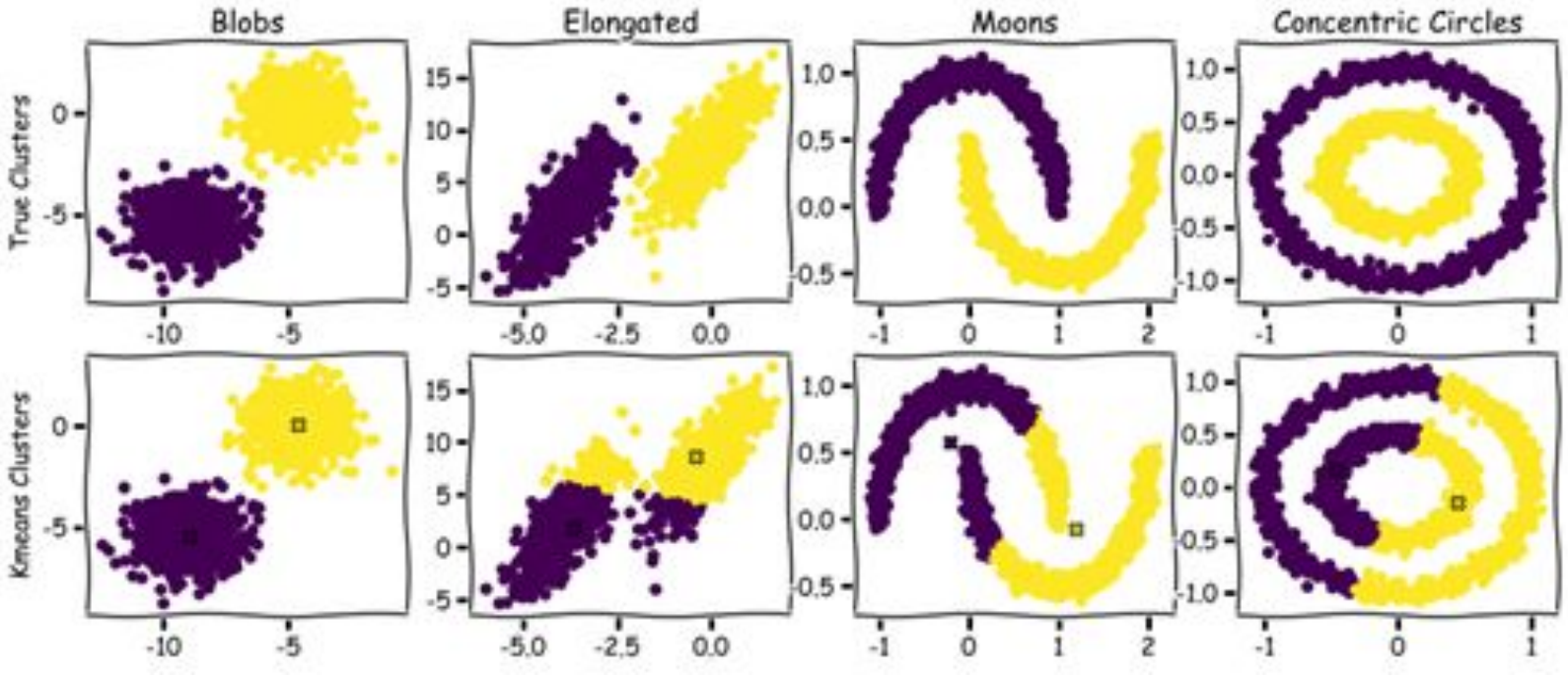
(You can determine this using the 'init' parameter in your model)

1. Default (`kmeans++`) – This initializes the centroids to be generally distant from each other
2. Randomly (`random`)
3. Manually (`ndarray`) – You can pass this argument and provide initial centers.

Note: Manual is recommended if you know your data well enough to see the clusters without much help, but rarely used in practice.

Flaws of K-Means Clustering

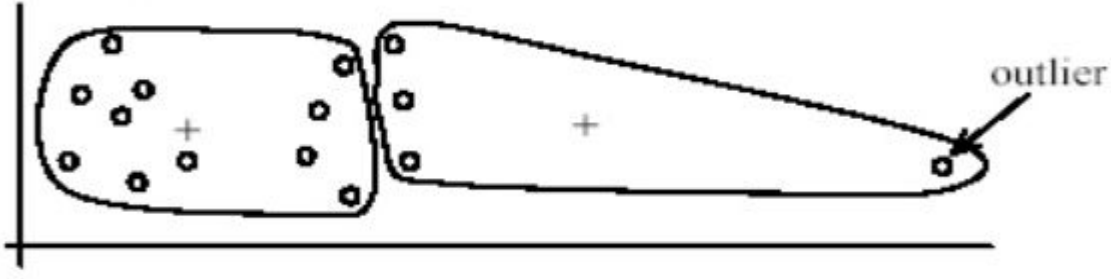
Fails with Non-Convex/Non-Round Shaped Clusters



Sensitive to Outliers

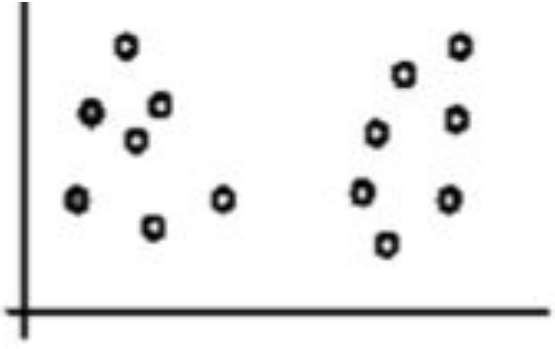


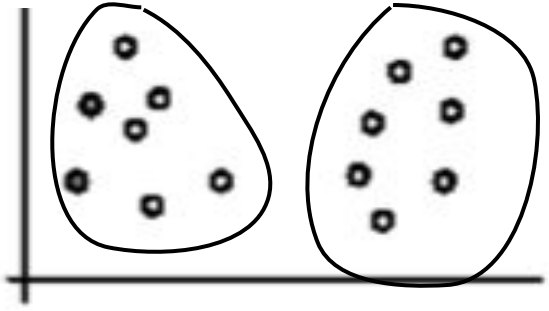
Ideal



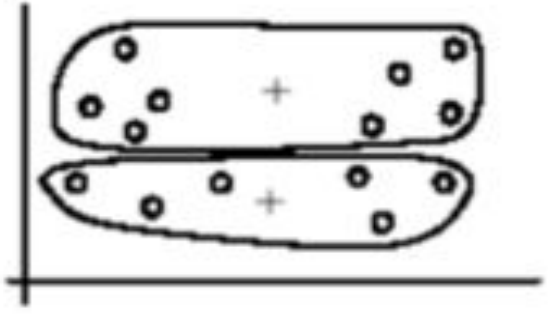
Less Desirable

Sensitive to Centroid Initialization



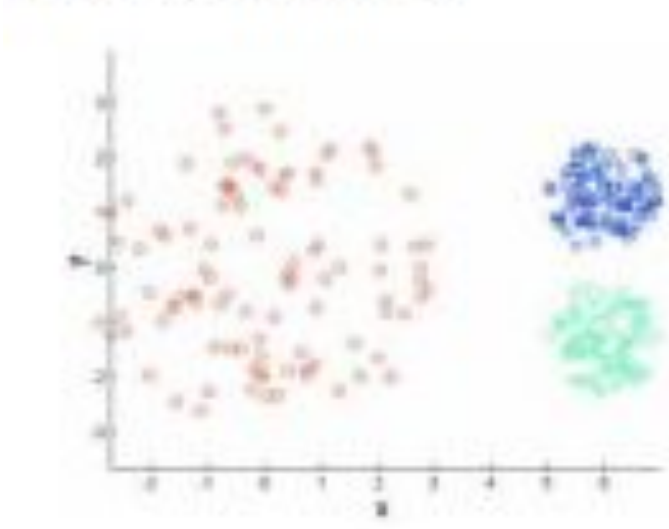


Ideal

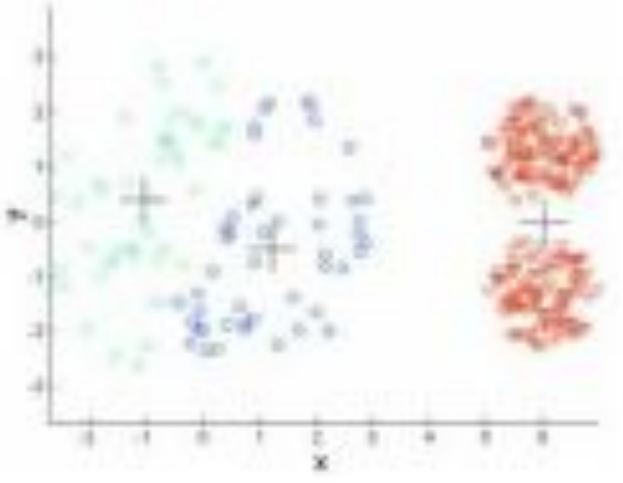


Less Desirable

Fails with Clusters of Different Densities



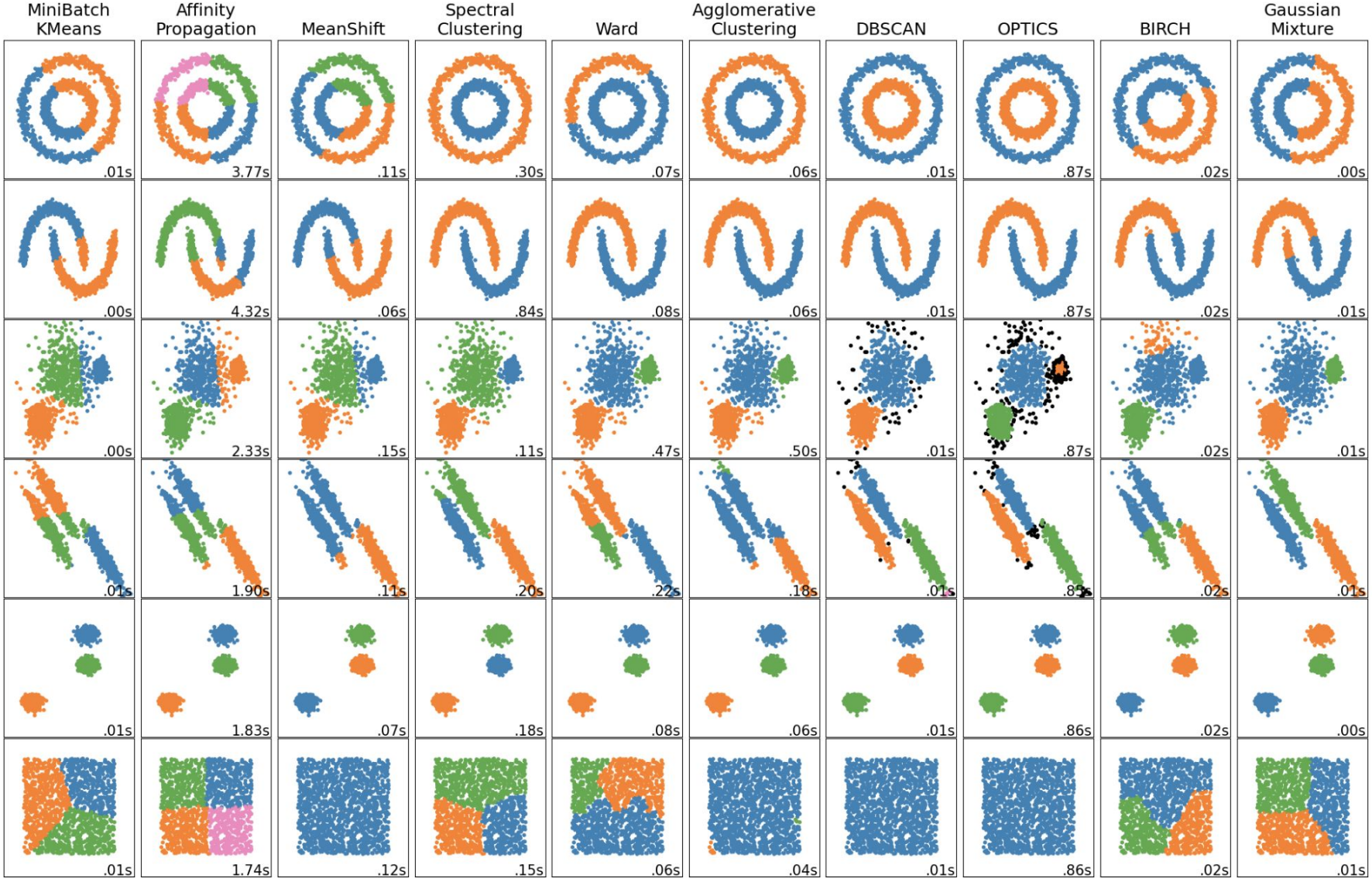
Ideal



Less Desirable

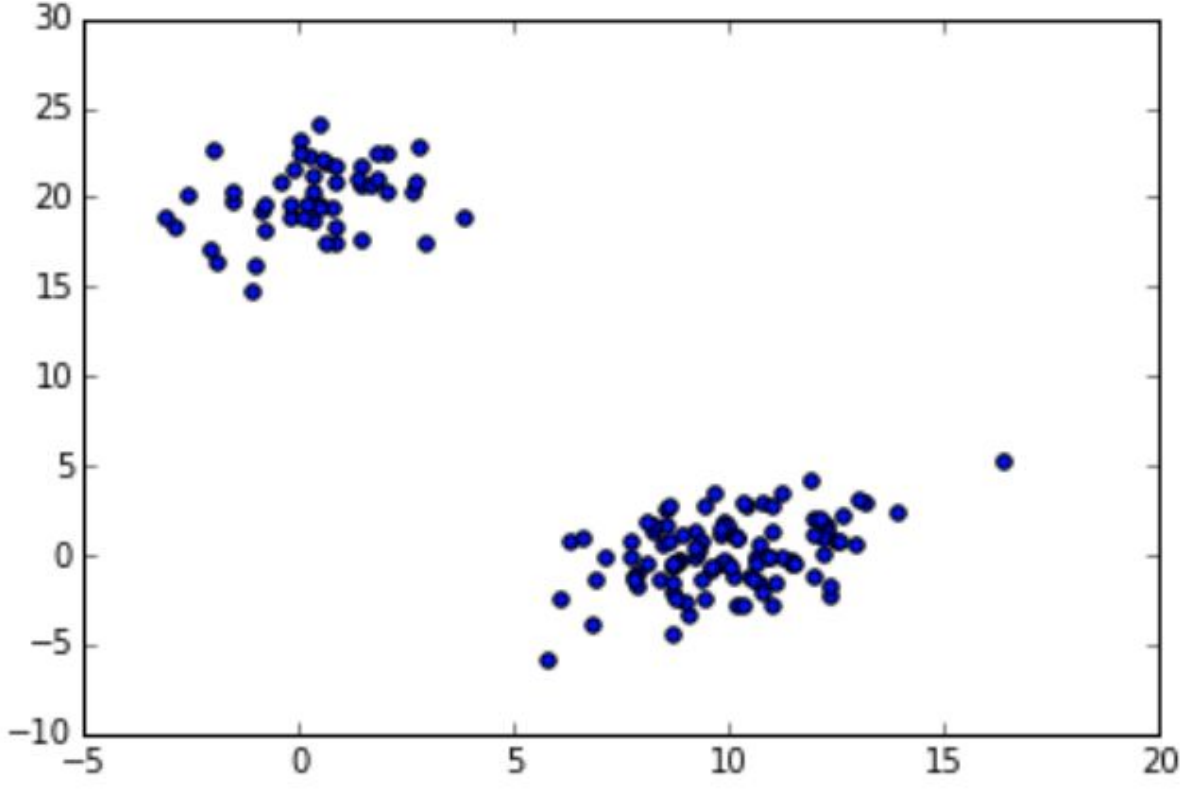
Other Clustering Models

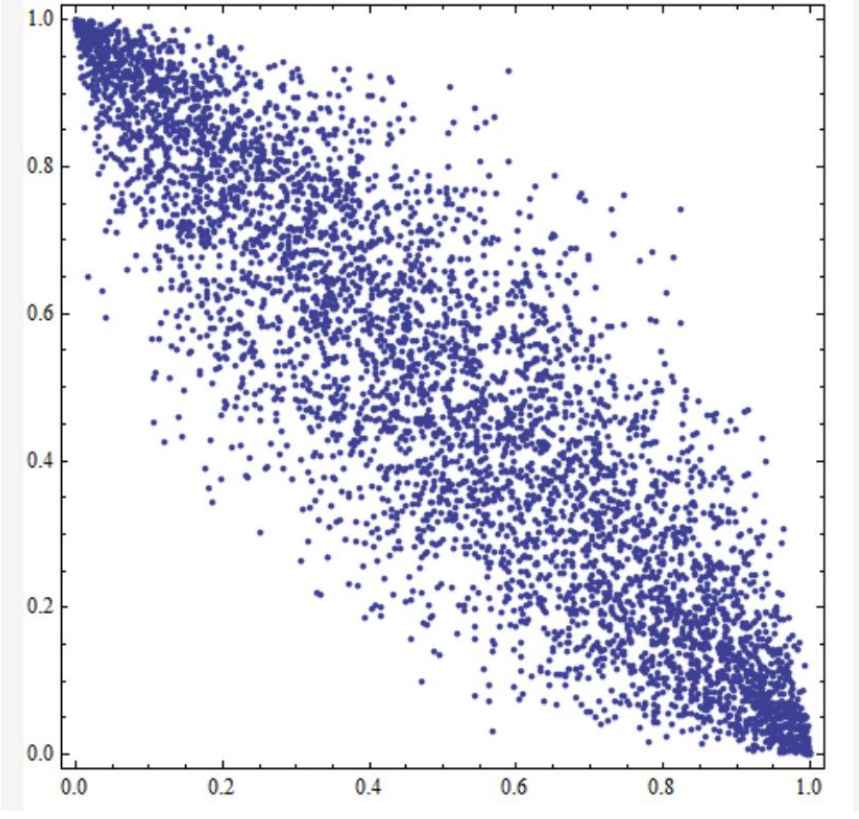


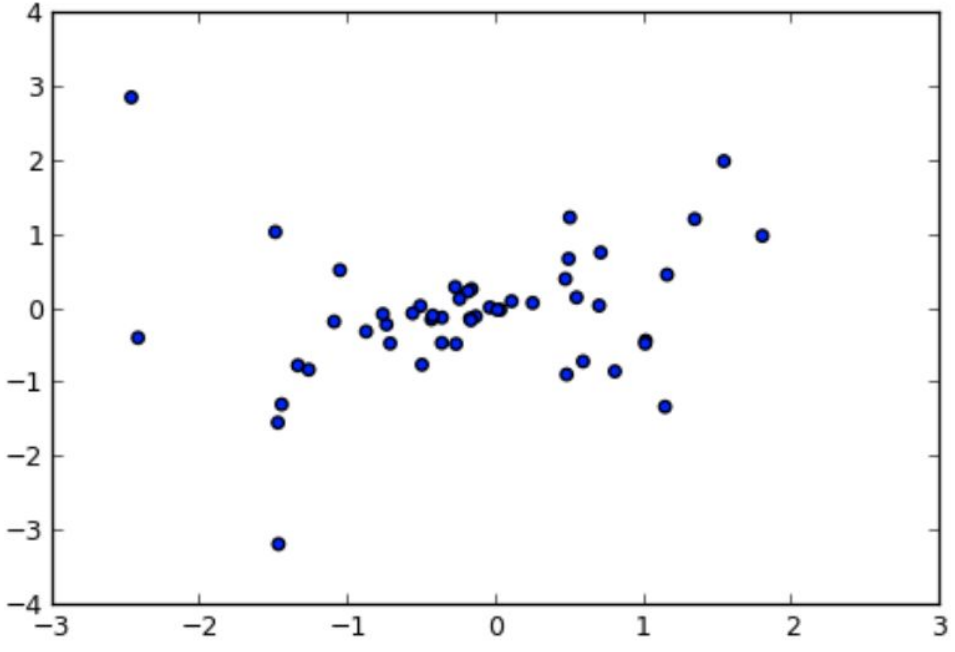

MiniBatch KMeans
Affinity Propagation
MeanShift
Spectral Clustering
Ward
Agglomerative Clustering
DBSCAN
OPTICS
BIRCH
Gaussian Mixture
.01s
3.77s
.11s
.30s
.07s
.06s
.01s
.87s
.02s
.00s
.00s
4.32s
.06s
.84s
.08s
.06s
.01s
.87s
.02s
.01s
.00s
2.33s
.15s
.11s
.47s
.50s
.01s
.87s
.02s
.01s
.01s
1.90s
.11s
.20s
.22s
.18s
.01s
.02s
.01s
.01s
1.83s
.07s
.18s
.08s
.06s
.01s
.86s
.02s
.00s
.01s
1.74s
.12s
.15s
.06s
.04s
.01s
.86s
.02s
.01s

Let's Play a Game...

How many clusters (k) should we choose?


30
25
20
15
10
5
0
−5
−10
−5
0
5
10
15
20


1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0


4
3
2
1
0
−1
−2
−3
−4
−3
−2
−1
0
1
2
3

In Slack:

Let’s brainstorm: How can we measure how “good” our clusters are?

Choosing *k*

This still remains an open question. After all, we're tuning a tuning parameter with no metric for success! Here are three ideas:

- Make an educated guess
  - Industry knowledge (there are five kinds of Democrats...)
  - Visualization (probably impossible if you have more than 2 variables)
- If you're using the resulting cluster labels as input to a supervised learning method later, you can tune k to have the best supervised learning model. This is **transfer learning**.
- Using the “elbow method” to find a number of clusters that no longer seems to improve a clustering metric by a noticeable degree.
- Judge based on a pseudo-evaluation metric, like the **silhouette score**.

Metrics to Evaluate Clustering

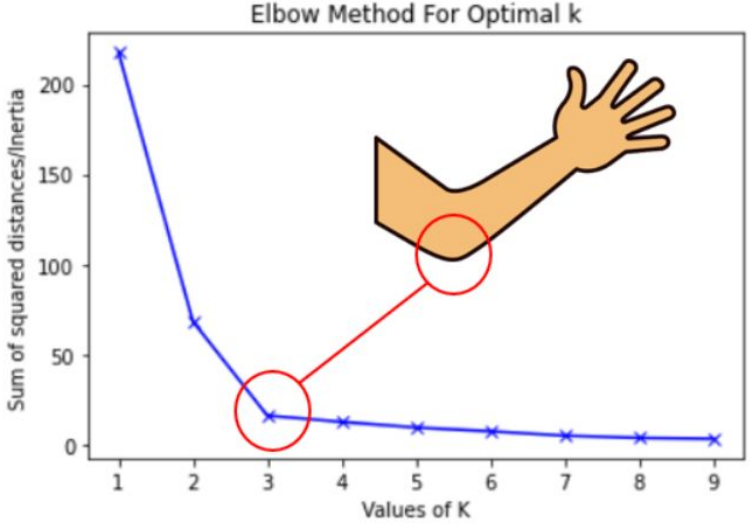
Inertia & Silhouette Score

Inertia - Sum of Squared Errors for each cluster

Low Inertia = Dense Cluster

Keep in Mind: As the number of clusters K increases, inertia will decrease. (If we add more clusters, the points in each cluster will become closer to each other).

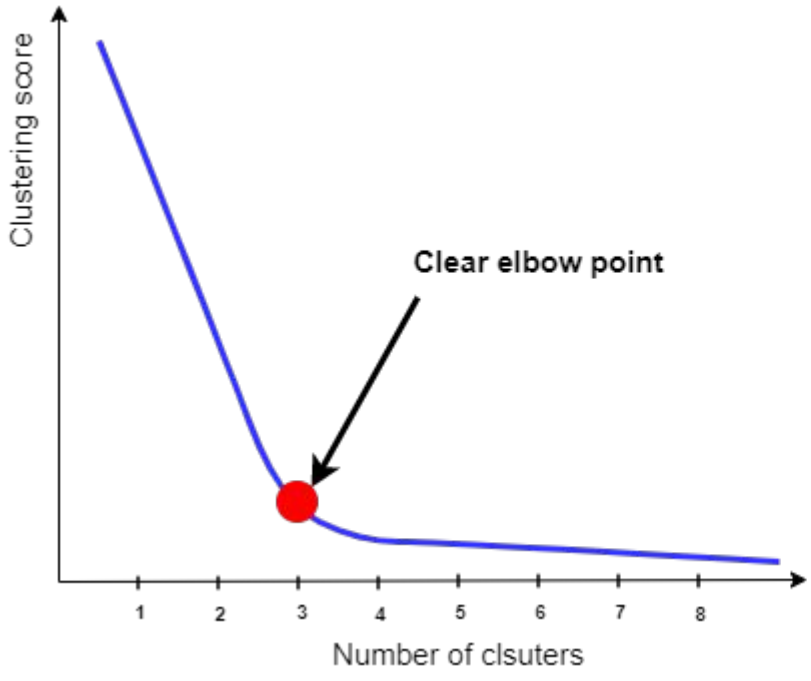
Therefore we want to identify the number of clusters that no longer seems to improve a clustering metric by a noticeable degree (the elbow)!

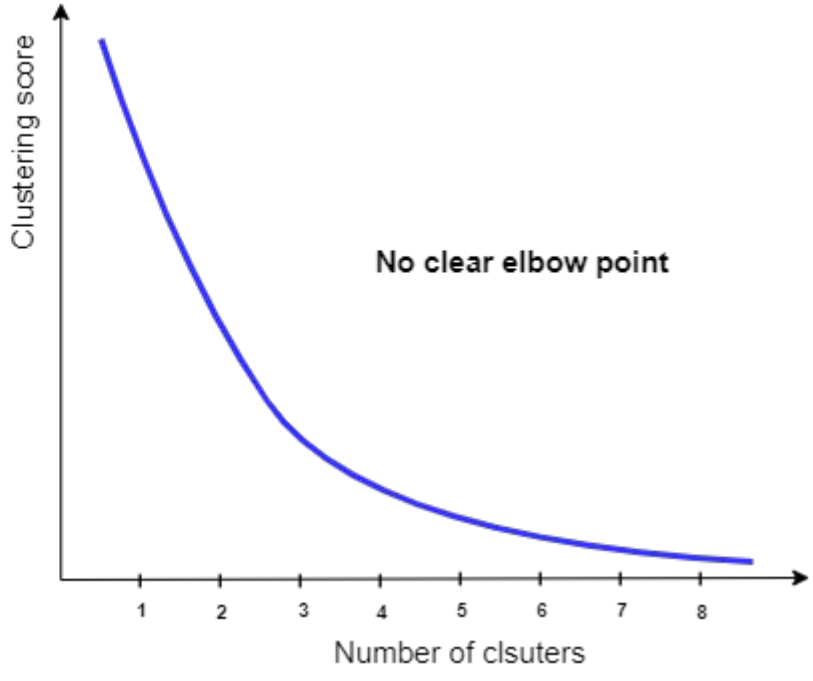


Line plot between K and inertia

 © 2018 General Assembly

Inertia - Sum of Squared Errors for each cluster





 © 2018 General Assembly

Silhouette Score - How Far Apart Clusters Are

On a scale of -1 to 1:

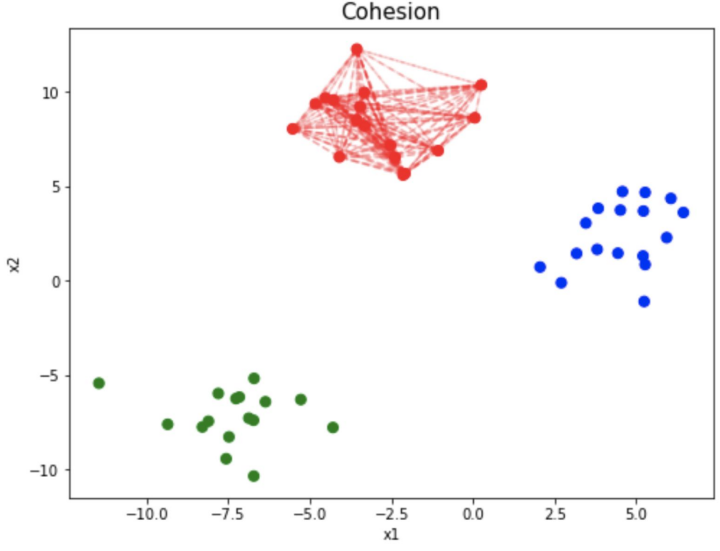
-1 = Bad

0 = Meh

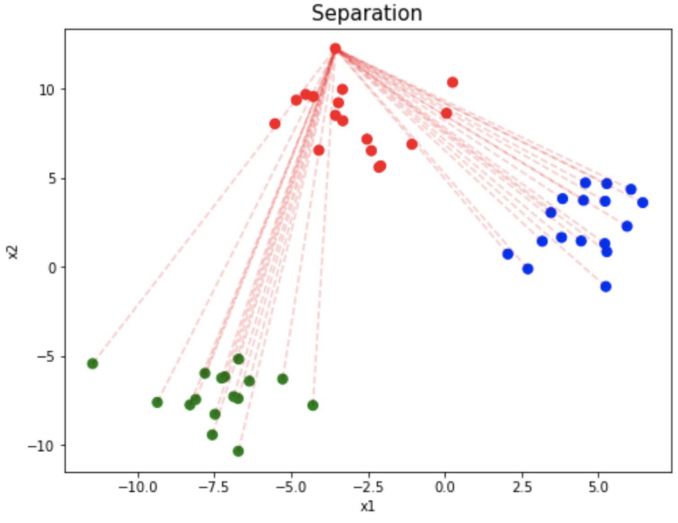
1 = Good

Silhouette Score - How is it Calculated?

Cohesion - Average Distance of Points within Clusters



Separation - Average Distance of Points in One Cluster to Points in Other Clusters



$$s(x) = \frac{separation(X) - cohesion(X)}{max\{cohesion(X), separation(X)\}}$$

$$SC(X) = \frac{1}{N} \sum_{n=1}^{N} s(x)$$

Remember, better clusters have a silhouette score closer to 1. Don't worry, sklearn has this built-in for us!


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